CN & NP

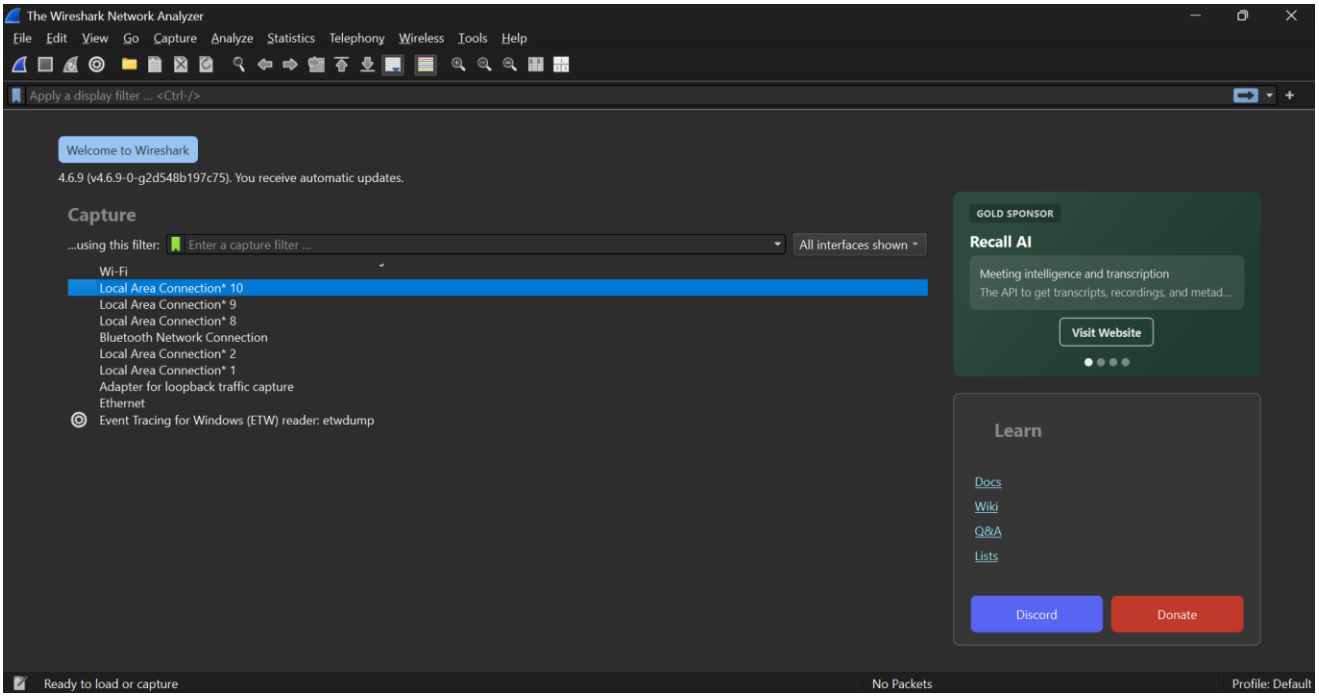
Name : M Ramya Sree **Roll No :** 2520030094

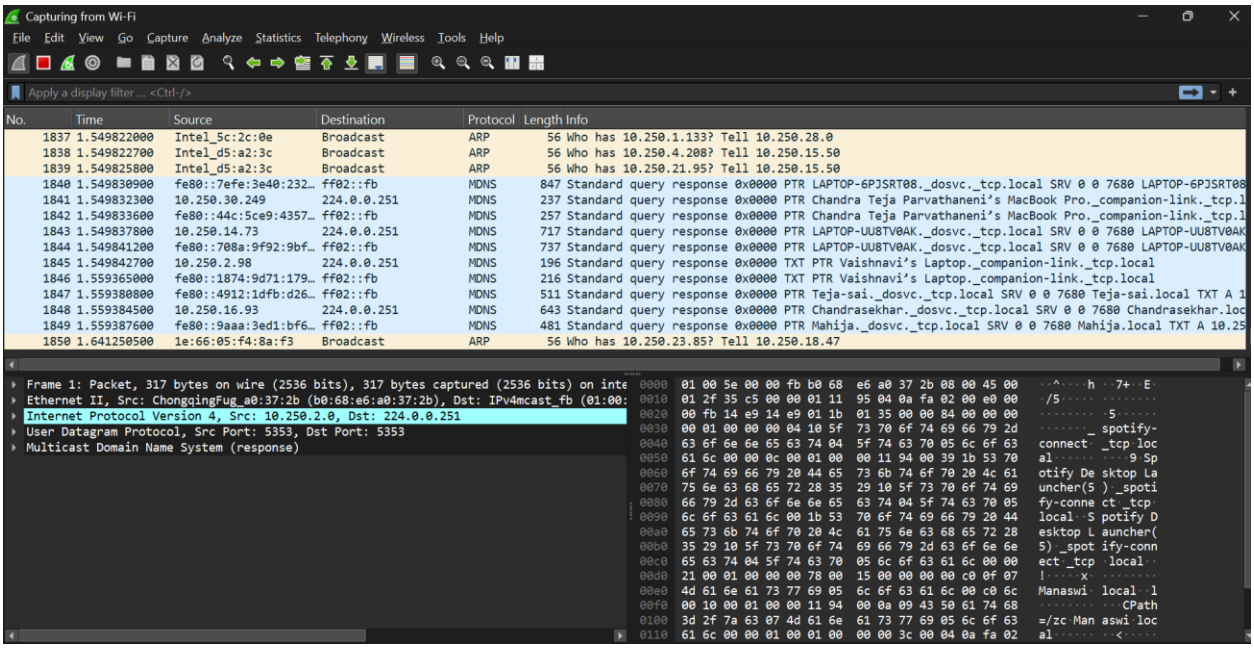
Skill – 1

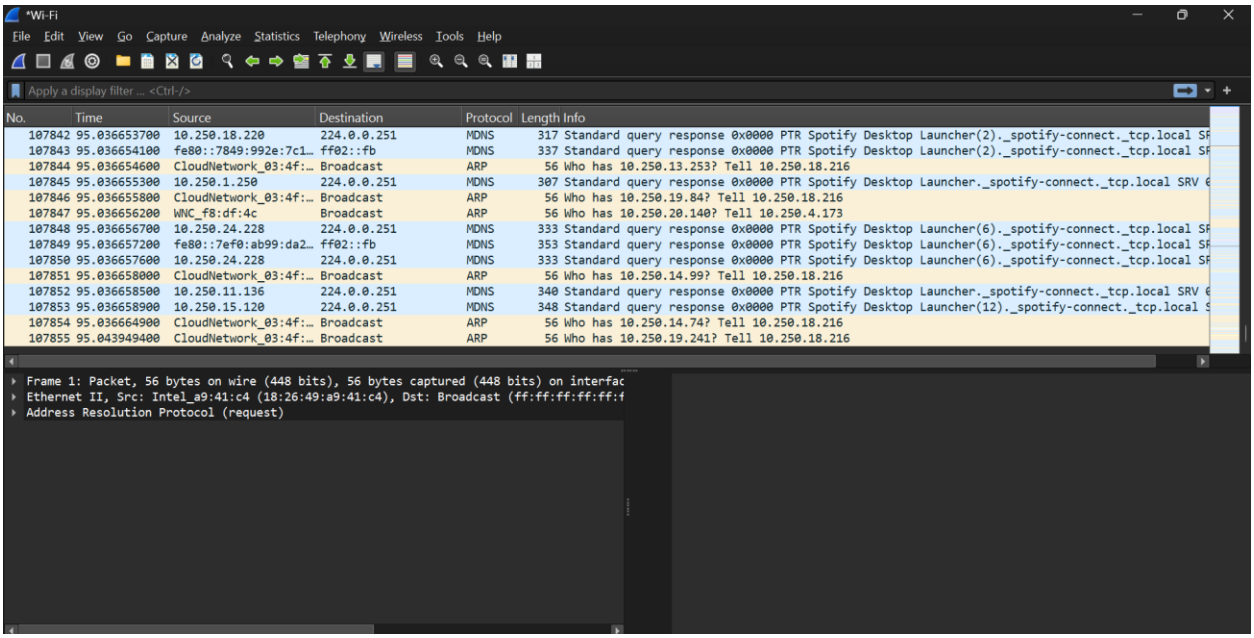
Encapsulation Builder : Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

Aim : To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission. Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead. Using Wireshark, we will capture real network packets, examine Ethernet, IP and TCP/UDP headers, calculate the total overhead, and determine how efficiently the network carries application data.

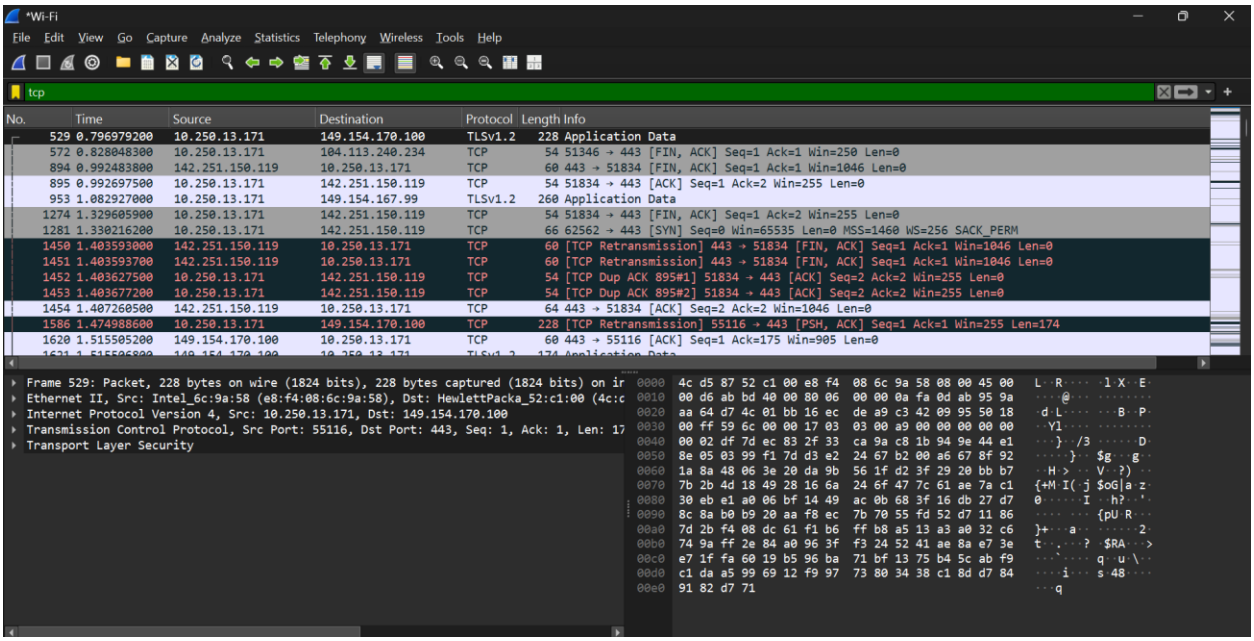
Screen Shots :



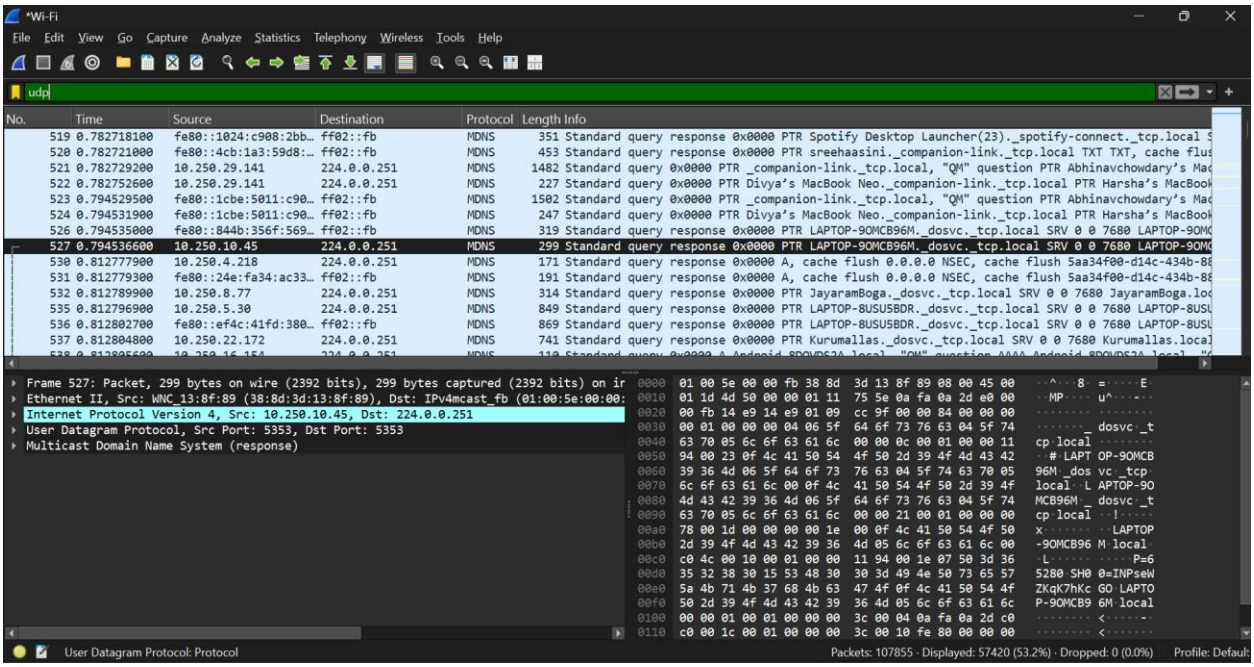

Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1837 1.549822000 Intel_5c:2c:0e Broadcast ARP 56 Who has 10.250.1.133? Tell 10.250.28.0
1838 1.549822700 Intel_d5:a2:3c Broadcast ARP 56 Who has 10.250.4.208? Tell 10.250.15.50
1839 1.549825800 Intel_d5:a2:3c Broadcast ARP 56 Who has 10.250.21.95? Tell 10.250.15.50
1840 1.549830900 fe80::7efe:3e40:232… ff02::fb MDNS 847 Standard query response 0x0000 PTR LAPTOP-6PJSRT08._dosvc._tcp.local SRV 0 0 7680 LAPTOP-6PJSRT08
1841 1.549832300 10.250.30.249 224.0.0.251 MDNS 237 Standard query response 0x0000 PTR Chandra Teja Parvathaneni's MacBook Pro._companion-link._tcp.l
1842 1.549833600 fe80::44c:5ce9:4357… ff02::fb MDNS 257 Standard query response 0x0000 PTR Chandra Teja Parvathaneni's MacBook Pro._companion-link._tcp.l
1843 1.549837800 10.250.14.73 224.0.0.251 MDNS 717 Standard query response 0x0000 PTR LAPTOP-UU8TV0AK._dosvc._tcp.local SRV 0 0 7680 LAPTOP-UU8TV0AK
1844 1.549841200 fe80::708a:9f92:9bf… ff02::fb MDNS 737 Standard query response 0x0000 PTR LAPTOP-UU8TV0AK._dosvc._tcp.local SRV 0 0 7680 LAPTOP-UU8TV0AK
1845 1.549842700 10.250.2.98 224.0.0.251 MDNS 196 Standard query response 0x0000 TXT PTR Vaishnavi's Laptop._companion-link._tcp.local
1846 1.559365000 fe80::1874:9d71:179… ff02::fb MDNS 216 Standard query response 0x0000 TXT PTR Vaishnavi's Laptop._companion-link._tcp.local
1847 1.559380800 fe80::4912:1dfb:d26… ff02::fb MDNS 511 Standard query response 0x0000 PTR Teja-sai._dosvc._tcp.local SRV 0 0 7680 Teja-sai.local TXT A 1
1848 1.559384500 10.250.16.93 224.0.0.251 MDNS 643 Standard query response 0x0000 PTR Chandrasekhar._dosvc._tcp.local SRV 0 0 7680 Chandrasekhar.loc
1849 1.559387600 fe80::9aaa:3ed1:bf6… ff02::fb MDNS 481 Standard query response 0x0000 PTR Mahija._dosvc._tcp.local SRV 0 0 7680 Mahija.local TXT A 10.25
1850 1.641250500 1e:66:05:f4:8a:f3 Broadcast ARP 56 Who has 10.250.23.85? Tell 10.250.18.47
Frame 1: Packet, 317 bytes on wire (2536 bits), 317 bytes captured (2536 bits) on inte
Ethernet II, Src: ChongqingFug_a0:37:2b (b0:68:e6:a0:37:2b), Dst: IPv4mcast_fb (01:00:
Internet Protocol Version 4, Src: 10.250.2.0, Dst: 224.0.0.251
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (response)
0000 01 00 5e 00 00 fb b0 68 e6 a0 37 2b 08 00 45 00
0010 01 2f 35 c5 00 00 01 11 95 04 0a fa 02 00 e0 00
0020 00 fb 14 e9 14 e9 01 1b 01 35 00 00 84 00 00 00
0030 00 01 00 00 00 04 10 5f 73 70 6f 74 69 66 79 2d
0040 63 6f 6e 6e 65 63 74 04 5f 74 63 70 05 6c 6f 63
0050 61 6c 00 00 0c 00 01 00 00 11 94 00 39 1b 53 70
0060 6f 74 69 66 79 20 44 65 73 6b 74 6f 70 20 4c 61
0070 75 6e 63 68 65 72 28 35 29 10 5f 73 70 6f 74 69
0080 66 79 2d 63 6f 6e 6e 65 63 74 04 5f 74 63 70 05
0090 6c 6f 63 61 6c 00 1b 53 70 6f 74 69 66 79 20 44
00a0 65 73 6b 74 6f 70 20 4c 61 75 6e 63 68 65 72 28
00b0 35 29 10 5f 73 70 6f 74 69 66 79 2d 63 6f 6e 6e
00c0 65 63 74 04 5f 74 63 70 05 6c 6f 63 61 6c 00 00
00d0 21 00 01 00 00 00 78 00 15 00 00 00 00 c0 0f 07
00e0 4d 61 6e 61 73 77 69 05 6c 6f 63 61 6c 00 c0 6c
00f0 00 10 00 01 00 00 11 94 00 0a 09 43 50 61 74 68
0100 3d 2f 7a 63 07 4d 61 6e 61 73 77 69 05 6c 6f 63
0110 61 6c 00 00 01 00 01 00 00 00 3c 00 04 0a fa 02


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
107842 95.036653700 10.250.18.220 224.0.0.251 MDNS 317 Standard query response 0x0000 PTR Spotify Desktop Launcher(2)._spotify-connect._tcp.local SR
107843 95.036654100 fe80::7849:992e:7c1… ff02::fb MDNS 337 Standard query response 0x0000 PTR Spotify Desktop Launcher(2)._spotify-connect._tcp.local SR
107844 95.036654600 CloudNetwork_03:4f:… Broadcast ARP 56 Who has 10.250.13.253? Tell 10.250.18.216
107845 95.036655300 10.250.1.250 224.0.0.251 MDNS 307 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV
107846 95.036655800 CloudNetwork_03:4f:… Broadcast ARP 56 Who has 10.250.19.84? Tell 10.250.18.216
107847 95.036656200 WNC_f8:df:4c Broadcast ARP 56 Who has 10.250.20.140? Tell 10.250.4.173
107848 95.036656700 10.250.24.228 224.0.0.251 MDNS 333 Standard query response 0x0000 PTR Spotify Desktop Launcher(6)._spotify-connect._tcp.local SR
107849 95.036657200 fe80::7ef0:ab99:da2… ff02::fb MDNS 353 Standard query response 0x0000 PTR Spotify Desktop Launcher(6)._spotify-connect._tcp.local SR
107850 95.036657600 10.250.24.228 224.0.0.251 MDNS 333 Standard query response 0x0000 PTR Spotify Desktop Launcher(6)._spotify-connect._tcp.local SR
107851 95.036658000 CloudNetwork_03:4f:… Broadcast ARP 56 Who has 10.250.14.99? Tell 10.250.18.216
107852 95.036658500 10.250.11.136 224.0.0.251 MDNS 340 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV
107853 95.036658900 10.250.15.120 224.0.0.251 MDNS 348 Standard query response 0x0000 PTR Spotify Desktop Launcher(12)._spotify-connect._tcp.local
107854 95.036664900 CloudNetwork_03:4f:… Broadcast ARP 56 Who has 10.250.14.74? Tell 10.250.18.216
107855 95.043949400 CloudNetwork_03:4f:… Broadcast ARP 56 Who has 10.250.19.241? Tell 10.250.18.216
Frame 1: Packet, 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interfac
Ethernet II, Src: Intel_a9:41:c4 (18:26:49:a9:41:c4), Dst: Broadcast (ff:ff:ff:ff:ff:f
Address Resolution Protocol (request)

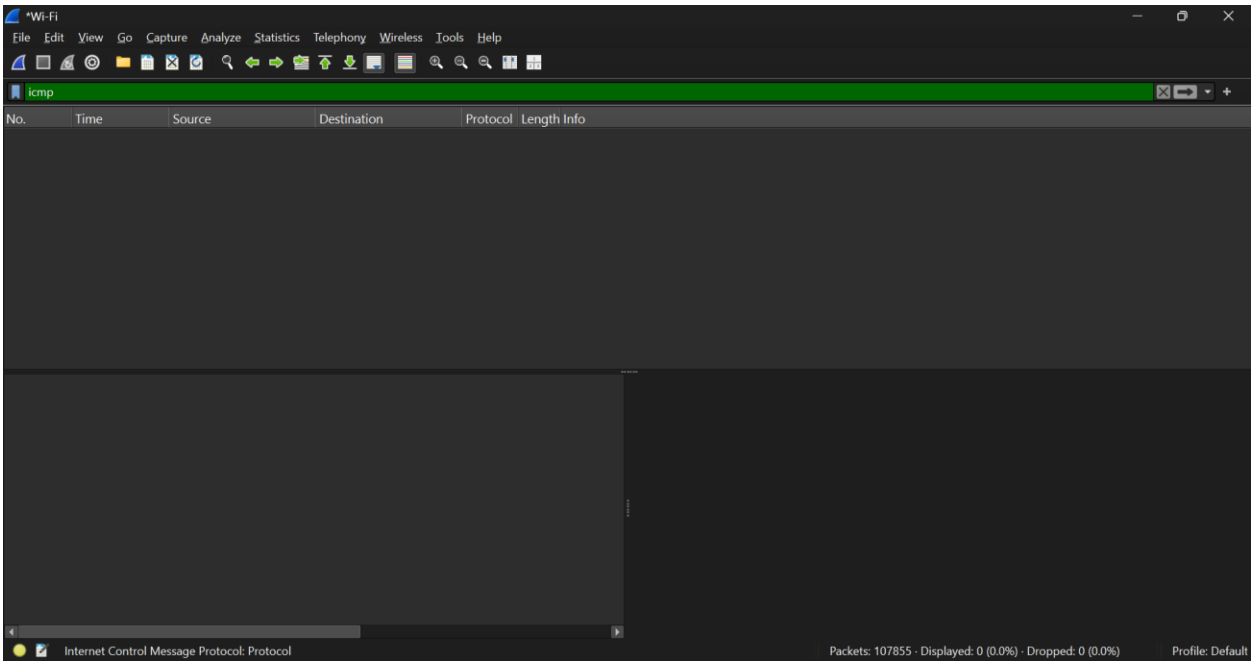
tcp :


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
529 0.796979200 10.250.13.171 149.154.170.100 TLSv1.2 228 Application Data
572 0.828048300 10.250.13.171 104.113.240.234 TCP 54 51346 → 443 [FIN, ACK] Seq=1 Ack=1 Win=250 Len=0
894 0.992483800 142.251.150.119 10.250.13.171 TCP 60 443 → 51834 [FIN, ACK] Seq=1 Ack=1 Win=1046 Len=0
895 0.992697500 10.250.13.171 142.251.150.119 TCP 54 51834 → 443 [ACK] Seq=1 Ack=2 Win=255 Len=0
953 1.082927000 10.250.13.171 149.154.167.99 TLSv1.2 260 Application Data
1274 1.329605900 10.250.13.171 142.251.150.119 TCP 54 51834 → 443 [FIN, ACK] Seq=1 Ack=2 Win=255 Len=0
1281 1.330216200 10.250.13.171 142.251.150.119 TCP 66 62562 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1450 1.403593000 142.251.150.119 10.250.13.171 TCP 60 [TCP Retransmission] 443 → 51834 [FIN, ACK] Seq=1 Ack=1 Win=1046 Len=0
1451 1.403593700 142.251.150.119 10.250.13.171 TCP 60 [TCP Retransmission] 443 → 51834 [FIN, ACK] Seq=1 Ack=1 Win=1046 Len=0
1452 1.403627500 10.250.13.171 142.251.150.119 TCP 54 [TCP Dup ACK 895#1] 51834 → 443 [ACK] Seq=2 Ack=2 Win=255 Len=0
1453 1.403677200 10.250.13.171 142.251.150.119 TCP 54 [TCP Dup ACK 895#2] 51834 → 443 [ACK] Seq=2 Ack=2 Win=255 Len=0
1454 1.407260500 142.251.150.119 10.250.13.171 TCP 64 443 → 51834 [ACK] Seq=2 Ack=2 Win=1046 Len=0
1586 1.474988600 10.250.13.171 149.154.170.100 TCP 228 [TCP Retransmission] 55116 → 443 [PSH, ACK] Seq=1 Ack=1 Win=255 Len=174
1620 1.515505200 149.154.170.100 10.250.13.171 TCP 60 443 → 55116 [ACK] Seq=1 Ack=175 Win=905 Len=0
Frame 529: Packet, 228 bytes on wire (1824 bits), 228 bytes captured (1824 bits) on ir
Ethernet II, Src: Intel_6c:9a:58 (e8:f4:08:6c:9a:58), Dst: HewlettPacka_52:c1:00 (4c:c
Internet Protocol Version 4, Src: 10.250.13.171, Dst: 149.154.170.100
Transmission Control Protocol, Src Port: 55116, Dst Port: 443, Seq: 1, Ack: 1, Len: 17
Transport Layer Security
0000 4c d5 87 52 c1 00 e8 f4 08 6c 9a 58 08 00 45 00
0010 00 d6 ab bd 40 00 80 06 00 00 0a fa 0d ab 95 9a
0020 aa 64 d7 4c 01 bb 16 ec de a9 c3 42 09 95 50 18
0030 00 ff 59 6c 00 00 17 03 03 00 a9 00 00 00 00 00
0040 00 02 df 7d ec 83 2f 33 ca 9a c8 1b 94 9e 44 e1
0050 8e 05 03 99 f1 7d d3 e2 24 67 b2 00 a6 67 8f 92
0060 1a 8a 48 06 3e 20 da 9b 56 1f d2 3f 29 20 bb b7
0070 7b 2b 4d 18 49 28 16 6a 24 6f 47 7c 61 ae 7a c1
0080 30 eb e1 a0 06 bf 14 49 ac 0b 68 3f 16 db 27 d7
0090 8c 8a b0 b9 20 aa f8 ec 7b 70 55 fd 52 d7 11 86
00a0 7d 2b f4 08 dc 61 f1 b6 ff b8 a5 13 a3 a0 32 c6
00b0 74 9a ff 2e 84 a0 96 3f f3 24 52 41 ae 8a e7 3e
00c0 e7 1f fa 60 19 b5 96 ba 71 bf 13 75 b4 5c ab f9
00d0 c1 da a5 99 69 12 f9 97 73 80 34 38 c1 8d d7 84
00e0 91 82 d7 71

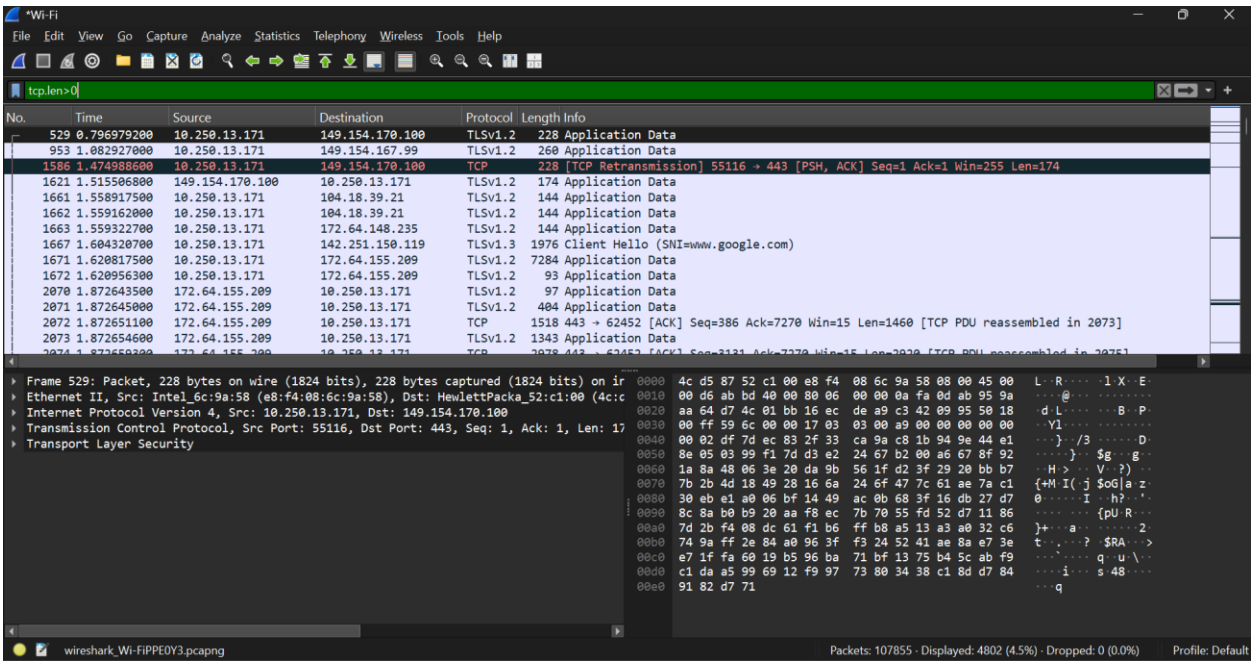
udp :

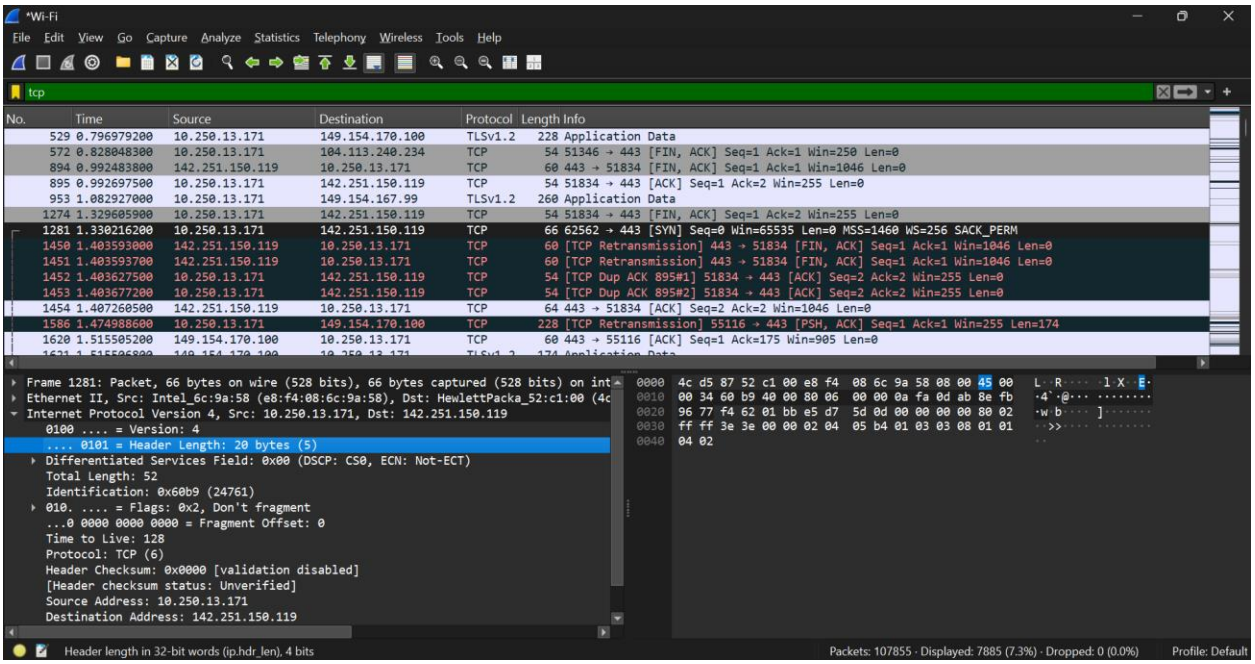

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Length Info
519 0.782718100 fe80::1024:c908:2bb… ff02::fb MDNS 351 Standard query response 0x0000 PTR Spotify Desktop Launcher(23)._spotify-connect._tcp.local
520 0.782721000 fe80::4cb:1a3:59d8:… ff02::fb MDNS 453 Standard query response 0x0000 PTR sreehaasini._companion-link._tcp.local TXT TXT, cache flus
521 0.782729200 10.250.29.141 224.0.0.251 MDNS 1482 Standard query 0x0000 PTR _companion-link._tcp.local, "QM" question PTR Abhinavchowdary's Mac
522 0.782752600 10.250.29.141 224.0.0.251 MDNS 227 Standard query 0x0000 PTR Divya's MacBook Neo._companion-link._tcp.local PTR Harsha's MacBook
523 0.794529500 fe80::1cbe:5011:c90… ff02::fb MDNS 1502 Standard query 0x0000 PTR _companion-link._tcp.local, "QM" question PTR Abhinavchowdary's Mac
524 0.794531900 fe80::1cbe:5011:c90… ff02::fb MDNS 247 Standard query 0x0000 PTR Divya's MacBook Neo._companion-link._tcp.local PTR Harsha's MacBook
526 0.794535000 fe80::844b:356f:569… ff02::fb MDNS 319 Standard query response 0x0000 PTR LAPTOP-9OMCB96M._dosvc._tcp.local SRV 0 0 7680 LAPTOP-9OMC
527 0.794536600 10.250.10.45 224.0.0.251 MDNS 299 Standard query response 0x0000 PTR LAPTOP-9OMCB96M._dosvc._tcp.local SRV 0 0 7680 LAPTOP-9OMC
530 0.812777900 10.250.4.218 224.0.0.251 MDNS 171 Standard query response 0x0000 A, cache flush 0.0.0.0 NSEC, cache flush 5aa34f00-d14c-434b-88
531 0.812779300 fe80::24e:fa34:ac33… ff02::fb MDNS 191 Standard query response 0x0000 A, cache flush 0.0.0.0 NSEC, cache flush 5aa34f00-d14c-434b-88
532 0.812789900 10.250.8.77 224.0.0.251 MDNS 314 Standard query response 0x0000 PTR JayaramBoga._dosvc._tcp.local SRV 0 0 7680 JayaramBoga.loc
535 0.812796900 10.250.5.30 224.0.0.251 MDNS 849 Standard query response 0x0000 PTR LAPTOP-8USU5BDR._dosvc._tcp.local SRV 0 0 7680 LAPTOP-8USU
536 0.812802700 fe80::ef4c:41fd:380… ff02::fb MDNS 869 Standard query response 0x0000 PTR LAPTOP-8USU5BDR._dosvc._tcp.local SRV 0 0 7680 LAPTOP-8USU
537 0.812804800 10.250.22.172 224.0.0.251 MDNS 741 Standard query response 0x0000 PTR Kurumallas._dosvc._tcp.local SRV 0 0 7680 Kurumallas.local
Frame 527: Packet, 299 bytes on wire (2392 bits), 299 bytes captured (2392 bits) on ir
Ethernet II, Src: WNC_13:8f:89 (38:8d:3d:13:8f:89), Dst: IPv4mcast_fb (01:00:5e:00:00:
Internet Protocol Version 4, Src: 10.250.10.45, Dst: 224.0.0.251
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (response)
0000 01 00 5e 00 00 fb 38 8d 3d 13 8f 89 08 00 45 00
0010 01 1d 4d 50 00 00 01 11 75 5e 0a fa 0a 2d e0 00
0020 00 fb 14 e9 14 e9 01 09 cc 9f 00 00 84 00 00 00
0030 00 01 00 00 00 04 06 5f 64 6f 73 76 63 04 5f 74
0040 63 70 05 6c 6f 63 61 6c 00 00 0c 00 01 00 00 11
0050 94 00 23 0f 4c 41 50 54 4f 50 2d 39 4f 4d 43 42
0060 39 36 4d 06 5f 64 6f 73 76 63 04 5f 74 63 70 05
0070 6c 6f 63 61 6c 00 0f 4c 41 50 54 4f 50 2d 39 4f
0080 4d 43 42 39 36 4d 06 5f 64 6f 73 76 63 04 5f 74
0090 63 70 05 6c 6f 63 61 6c 00 00 21 00 01 00 00 00
00a0 78 00 1d 00 00 00 00 1e 00 0f 4c 41 50 54 4f 50
00b0 2d 39 4f 4d 43 42 39 36 4d 05 6c 6f 63 61 6c 00
00c0 c0 4c 00 10 00 01 00 00 11 94 00 1e 07 50 3d 36
00d0 35 32 38 30 15 53 48 30 30 3d 49 4e 50 73 65 57
00e0 5a 4b 71 4b 37 68 4b 63 47 4f 0f 4c 41 50 54 4f
00f0 50 2d 39 4f 4d 43 42 39 36 4d 05 6c 6f 63 61 6c
0100 00 00 01 00 01 00 00 00 3c 00 04 0a fa 0a 2d c0
0110 c0 00 1c 00 01 00 00 00 3c 00 10 fe 80 00 00 00
User Datagram Protocol: Protocol
Packets: 107855 · Displayed: 57420 (53.2%) · Dropped: 0 (0.0%)
Profile: Default

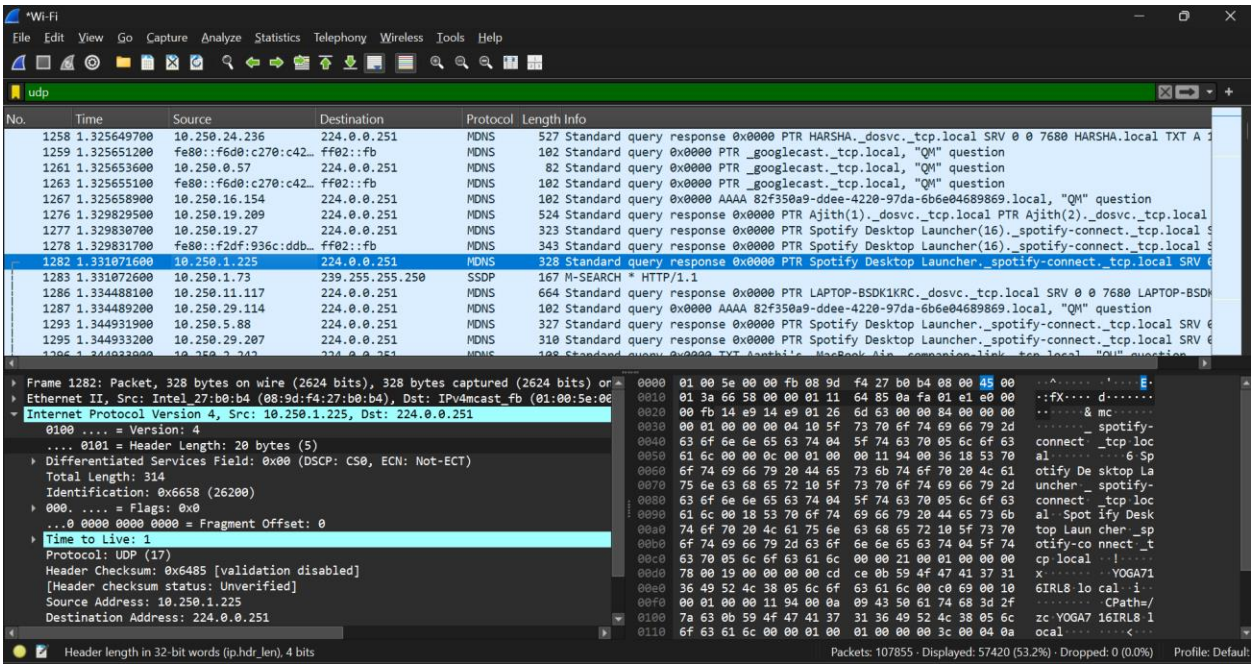
icmp :



tcp.len > 0 :




*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
529 0.796979200 10.250.13.171 149.154.170.100 TLSv1.2 228 Application Data
572 0.828048300 10.250.13.171 104.113.240.234 TCP 54 51346 → 443 [FIN, ACK] Seq=1 Ack=1 Win=250 Len=0
894 0.992483800 142.251.150.119 10.250.13.171 TCP 60 443 → 51834 [FIN, ACK] Seq=1 Ack=1 Win=1046 Len=0
895 0.992697500 10.250.13.171 142.251.150.119 TCP 54 51834 → 443 [ACK] Seq=1 Ack=2 Win=255 Len=0
953 1.082927000 10.250.13.171 149.154.167.99 TLSv1.2 260 Application Data
1274 1.329605900 10.250.13.171 142.251.150.119 TCP 54 51834 → 443 [FIN, ACK] Seq=1 Ack=2 Win=255 Len=0
1281 1.330216200 10.250.13.171 142.251.150.119 TCP 66 62562 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1450 1.403593000 142.251.150.119 10.250.13.171 TCP 60 [TCP Retransmission] 443 → 51834 [FIN, ACK] Seq=1 Ack=1 Win=1046 Len=0
1451 1.403593700 142.251.150.119 10.250.13.171 TCP 60 [TCP Retransmission] 443 → 51834 [FIN, ACK] Seq=1 Ack=1 Win=1046 Len=0
1452 1.403627500 10.250.13.171 142.251.150.119 TCP 54 [TCP Dup ACK 895#1] 51834 → 443 [ACK] Seq=2 Ack=2 Win=255 Len=0
1453 1.403677200 10.250.13.171 142.251.150.119 TCP 54 [TCP Dup ACK 895#2] 51834 → 443 [ACK] Seq=2 Ack=2 Win=255 Len=0
1454 1.407260500 142.251.150.119 10.250.13.171 TCP 64 443 → 51834 [ACK] Seq=2 Ack=2 Win=1046 Len=0
1586 1.474988600 10.250.13.171 149.154.170.100 TCP 228 [TCP Retransmission] 55116 → 443 [PSH, ACK] Seq=1 Ack=1 Win=255 Len=174
1620 1.515505200 149.154.170.100 10.250.13.171 TCP 60 443 → 55116 [ACK] Seq=1 Ack=175 Win=905 Len=0
Frame 1281: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on int
Ethernet II, Src: Intel_6c:9a:58 (e8:f4:08:6c:9a:58), Dst: HewlettPacka_52:c1:00 (4c
Internet Protocol Version 4, Src: 10.250.13.171, Dst: 142.251.150.119
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 52
Identification: 0x60b9 (24761)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.13.171
Destination Address: 142.251.150.119
0000 4c d5 87 52 c1 00 e8 f4 08 6c 9a 58 08 00 45 00
0010 00 34 60 b9 40 00 80 06 00 00 0a fa 0d ab 8e fb
0020 96 77 f4 62 01 bb e5 d7 5d 0d 00 00 00 00 80 02
0030 ff ff 3e 3e 00 00 02 04 05 b4 01 03 03 08 01 01
0040 04 02
Header length in 32-bit words (ip.hdr_len), 4 bits
Packets: 107855 · Displayed: 7885 (7.3%) · Dropped: 0 (0.0%)
Profile: Default


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Length Info
1258 1.325649700 10.250.24.236 224.0.0.251 MDNS 527 Standard query response 0x0000 PTR HARSHA._dosvc._tcp.local SRV 0 0 7680 HARSHA.local TXT A 1
1259 1.325651200 fe80::f6d0:c270:c42… ff02::fb MDNS 102 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
1261 1.325653600 10.250.0.57 224.0.0.251 MDNS 82 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
1263 1.325655100 fe80::f6d0:c270:c42… ff02::fb MDNS 102 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
1267 1.325658900 10.250.16.154 224.0.0.251 MDNS 102 Standard query 0x0000 AAAA 82f350a9-ddee-4220-97da-6b6e04689869.local, "QM" question
1276 1.329829500 10.250.19.209 224.0.0.251 MDNS 524 Standard query response 0x0000 PTR Ajith(1)._dosvc._tcp.local PTR Ajith(2)._dosvc._tcp.local
1277 1.329830700 10.250.19.27 224.0.0.251 MDNS 323 Standard query response 0x0000 PTR Spotify Desktop Launcher(16)._spotify-connect._tcp.local S
1278 1.329831700 fe80::f2df:936c:ddb… ff02::fb MDNS 343 Standard query response 0x0000 PTR Spotify Desktop Launcher(16)._spotify-connect._tcp.local S
1282 1.331071600 10.250.1.225 224.0.0.251 MDNS 328 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0
1283 1.331072600 10.250.1.73 239.255.255.250 SSDP 167 M-SEARCH * HTTP/1.1
1286 1.334488100 10.250.11.117 224.0.0.251 MDNS 664 Standard query response 0x0000 PTR LAPTOP-BSDK1KRC._dosvc._tcp.local SRV 0 0 7680 LAPTOP-BSDK
1287 1.334489200 10.250.29.114 224.0.0.251 MDNS 102 Standard query 0x0000 AAAA 82f350a9-ddee-4220-97da-6b6e04689869.local, "QM" question
1293 1.344931900 10.250.5.88 224.0.0.251 MDNS 327 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0
1295 1.344933200 10.250.29.207 224.0.0.251 MDNS 310 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0
Frame 1282: Packet, 328 bytes on wire (2624 bits), 328 bytes captured (2624 bits) or
Ethernet II, Src: Intel_27:b0:b4 (08:9d:f4:27:b0:b4), Dst: IPv4mcast_fb (01:00:5e:0
Internet Protocol Version 4, Src: 10.250.1.225, Dst: 224.0.0.251
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 314
Identification: 0x6658 (26200)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 1
Protocol: UDP (17)
Header Checksum: 0x6485 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.1.225
Destination Address: 224.0.0.251
0000 01 00 5e 00 00 fb 08 9d f4 27 b0 b4 08 00 45 00
0010 01 3a 66 58 00 00 01 11 64 85 0a fa 01 e1 e0 00
0020 00 fb 14 e9 14 e9 01 26 6d 63 00 00 84 00 00 00
0030 00 01 00 00 00 04 10 5f 73 70 6f 74 69 66 79 2d
0040 63 6f 6e 6e 65 63 74 04 5f 74 63 70 05 6c 6f 63
0050 61 6c 00 00 0c 00 01 00 00 11 94 00 36 18 53 70
0060 6f 74 69 66 79 20 44 65 73 6b 74 6f 70 20 4c 61
0070 75 6e 63 68 65 72 10 5f 73 70 6f 74 69 66 79 2d
0080 63 6f 6e 6e 65 63 74 04 5f 74 63 70 05 6c 6f 63
0090 61 6c 00 18 53 70 6f 74 69 66 79 20 44 65 73 6b
00a0 74 6f 70 20 4c 61 75 6e 63 68 65 72 10 5f 73 70
00b0 6f 74 69 66 79 2d 63 6f 6e 6e 65 63 74 04 5f 74
00c0 63 70 05 6c 6f 63 61 6c 00 00 21 00 01 00 00 00
00d0 78 00 19 00 00 00 00 cd ce 0b 59 4f 47 41 37 31
00e0 36 49 52 4c 38 05 6c 6f 63 61 6c 00 c0 69 00 10
00f0 00 01 00 00 11 94 00 0a 09 43 50 61 74 68 3d 2f
0100 7a 63 0b 59 4f 47 41 37 31 36 49 52 4c 38 05 6c
0110 6f 63 61 6c 00 00 01 00 01 00 00 00 3c 00 04 0a
Header length in 32-bit words (ip.hdr_len), 4 bits
Packets: 107855 · Displayed: 57420 (53.2%) · Dropped: 0 (0.0%)
Profile: Default

Step – 6 :

Parameter	TCP Packet	UDP Packet
Frame Length	54 bytes	328 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	20 bytes	8 bytes
IP Total Length	40 bytes	314 bytes
Transport Segment/Datagram Length	20 bytes	294 bytes
Payload Length	0 bytes	286 bytes